

Solutions

Numbering follows the lecture notes. For TYPE C exercises the idea is boxed; the rest are routine once the idea is in hand, which is what the types mean.

Lectures 5–8

§1.3 The Gaussian Wigner matrix

1.9 A $M_{ij} = \frac{1}{\sqrt{2N}}(A_{ij} + A_{ji})$ with the A_{ab} i.i.d. standard, so $\mathbb{E}[A_{ab}A_{cd}] = \delta_{ac}\delta_{bd}$ and

$$\mathbb{E}[M_{ij}M_{kl}] = \frac{1}{2N}\mathbb{E}[(A_{ij} + A_{ji})(A_{kl} + A_{lk})] = \frac{1}{2N}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} + \delta_{il}\delta_{jk} + \delta_{ik}\delta_{jl}),$$

which is $\frac{1}{N}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$. At $i = j = k = l$ both delta terms fire and the formula returns $\frac{2}{N}$, so the diagonal variance is not a convention imposed on the model but a consequence of symmetrizing A : the diagonal entry $M_{ii} = \frac{1}{\sqrt{2N}}(A_{ii} + A_{ii})$ collects the same Gaussian twice.

§1.5* Cumulants

1.14 B Write $K(t) = \log \mathbb{E}[e^{\sum_i t_i Y_i}]$.

(a) Symmetry is the equality of mixed partials. For linearity let $\tilde{K}(s, s', t_2, \dots, t_m) = \log \mathbb{E} \exp(sY + s'Z + \sum_{i \geq 2} t_i Y_i)$. The log-transform of $(aY + bZ, Y_2, \dots, Y_m)$ is $K(t_1, \dots, t_m) = \tilde{K}(at_1, bt_1, t_2, \dots, t_m)$, so by the chain rule, applied once since t_1 is differentiated once,

$$\partial_{t_1} K = a \partial_s \tilde{K} + b \partial_{s'} \tilde{K}.$$

Now apply $\partial_{t_2} \cdots \partial_{t_m}$ and set every variable to 0: the two terms are $a \kappa(Y, Y_2, \dots)$ and $b \kappa(Z, Y_2, \dots)$. Adding a constant c to Y_1 adds ct_1 to K , which any ∂_{t_j} with $j \geq 2$ kills. Applying linearity in each of the m slots gives $\kappa(cY, \dots, cY) = c^m \kappa(Y, \dots, Y)$.

(b) If the arguments split into independent groups $B_1 \sqcup B_2$, then $\mathbb{E}[e^{\sum t_i Y_i}]$ factors and $K = F_1(t_{B_1}) + F_2(t_{B_2})$. The derivative $\partial_{t_1} \cdots \partial_{t_m}$ contains a derivative in each group, so it kills both summands. For additivity expand $\kappa(Y + Z, \dots, Y + Z)$ by (a) into 2^m terms $\kappa(W_1, \dots, W_m)$ with each $W_j \in \{Y, Z\}$; every term containing both Y and Z vanishes, leaving $\kappa(Y, \dots, Y) + \kappa(Z, \dots, Z)$.

(c) Definition 1.3 of Lectures 1–2 at $\xi = it$ gives $K(t) = \frac{1}{2}t^\top \Sigma t$, a quadratic form: its first derivatives vanish at 0, its second are Σ_{ab} , and all higher ones are identically 0.

1.15 B

Multilinearity expands $\kappa(S_n, \dots, S_n)$ into n^m joint cumulants of the X_i , and independence **kills every one that is not on the diagonal**.

By Exercise 1.14(a) we may drop the centering and pull $n^{-1/2}$ out of each of the m slots:

$$\kappa(S_n, \dots, S_n) = n^{-m/2} \sum_{i_1, \dots, i_m=1}^n \kappa(X_{i_1}, \dots, X_{i_m}).$$

If the indices are not all equal, the arguments split into those equal to X_{i_1} and the rest, a non-empty group independent of the first, so that term vanishes by Exercise 1.14(b). The n diagonal terms each equal $\kappa(X_1, \dots, X_1)$, so

$$\kappa(S_n, \dots, S_n) = n \cdot n^{-m/2} \kappa(X_1, \dots, X_1) = n^{1-m/2} \kappa(X_1, \dots, X_1).$$

At $m = 2$ the exponent is 0 and $\kappa(S_n, S_n) = \kappa(X_1, X_1) = \sigma^2$ for every n . For $m \geq 3$ it is negative, so every higher cumulant tends to 0: in the limit only the second cumulant survives, which is exactly what a Gaussian looks like.

§2.3 Two moments by hand

2.6 A $\text{tr } M^{2k+1}$ is a sum of products of an odd number of centered jointly Gaussian variables, and Theorem 1.2 pairs factors two at a time, so every term has mean 0. (Alternatively, $-M$ has the same law as M , so $\text{tr } M^{2k+1}$ is a symmetric random variable.) Next, $\text{tr } M = \sum_i M_{ii}$ is a sum of N independent centered variables of variance $\frac{2}{N}$, so $\mathbb{E}[(\text{tr } M)^2] = 2$.

§2.4–§2.7 Paths, every moment, and the fluctuations

2.20 A Of the $5!! = 15$ pair partitions of $\{1, \dots, 6\}$, ten contain a crossing pair (for example $\{13\}\{24\}\{56\}$, crossing at $1 < 2 < 3 < 4$), and five are non-crossing. For each of those five, a tree path with that pairing, starting from a vertex r :

$\{12\}\{34\}\{56\}$	$r \rightarrow a \rightarrow r \rightarrow b \rightarrow r \rightarrow c \rightarrow r$	three edges from r
$\{12\}\{36\}\{45\}$	$r \rightarrow a \rightarrow r \rightarrow b \rightarrow c \rightarrow b \rightarrow r$	one edge, then a path of two
$\{14\}\{23\}\{56\}$	$r \rightarrow a \rightarrow b \rightarrow a \rightarrow r \rightarrow c \rightarrow r$	a path of two, then one edge
$\{16\}\{23\}\{45\}$	$r \rightarrow a \rightarrow b \rightarrow a \rightarrow c \rightarrow a \rightarrow r$	one edge, then two excursions from its end
$\{16\}\{25\}\{34\}$	$r \rightarrow a \rightarrow b \rightarrow c \rightarrow b \rightarrow a \rightarrow r$	a path of three, out and back

Each visits r, a, b, c : four vertices, $k+1$ for $k=3$.

2.21 B ρ_{sc} is even, so its odd moments vanish. Substituting $x = 2 \sin \theta$ gives $\sqrt{4-x^2} = 2 \cos \theta$ and $dx = 2 \cos \theta d\theta$ on $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, so

$$\int_{-2}^2 x^{2k} \rho_{sc}(x) dx = \frac{4^{k+1}}{2\pi} \int_{-\pi/2}^{\pi/2} \sin^{2k} \theta \cos^2 \theta d\theta.$$

At $k=0$ the integral is $\int \cos^2 = \frac{\pi}{2}$, giving $\frac{4}{2\pi} \cdot \frac{\pi}{2} = 1$: ρ_{sc} is a probability density. At $k=1$, $\int_{-\pi/2}^{\pi/2} \sin^2 \theta \cos^2 \theta d\theta = \frac{1}{8} \int_{-\pi/2}^{\pi/2} \sin^2 2\theta d\theta = \frac{\pi}{8}$, giving $\frac{16}{2\pi} \cdot \frac{\pi}{8} = 1 = C_1$. At $k=2$, $\int_{-\pi/2}^{\pi/2} \sin^4 \theta \cos^2 \theta d\theta = \frac{\pi}{16}$, giving $\frac{64}{2\pi} \cdot \frac{\pi}{16} = 2 = C_2$.

§3 The Stieltjes transform

3.11 A (a) $M=0$: every $\lambda_i = 0$, so $m_N(z) = -\frac{1}{z}$. (b) Half at ± 1 :

$$m_N(z) = \frac{1}{2} \left(\frac{1}{-1-z} + \frac{1}{1-z} \right) = \frac{-z}{z^2-1}.$$

In both cases $|\lambda_i - z| \geq \Im z$ for real λ_i , so $|m_N(z)| \leq \frac{1}{\Im z}$, which is (7). And $\frac{1}{\lambda-z} = -\frac{1}{z} \left(1 - \frac{\lambda}{z}\right)^{-1} = -\frac{1}{z} - \frac{\lambda}{z^2} - \dots$, averaged over i , gives $m_N(z) = -\frac{1}{z} - \frac{\text{tr } M/N}{z^2} + O(|z|^{-3})$; here $\text{tr } M = 0$ in both cases, so the second term drops out.

3.12 A At $\varepsilon = 1$ the curve is a broad bump — the semicircle smoothed by a Cauchy kernel of width 1, too smooth to show the edges at ± 2 . At $\varepsilon = 0.1$ it tracks ρ_{sc} closely, edges included: the kernel still averages over $\varepsilon N \approx 200$ eigenvalues. At $\varepsilon = 0.001$ the individual eigenvalues resolve into spikes of height $\sim \frac{1}{\pi N \varepsilon}$. The crossover is at $\varepsilon \sim 1/N$, the mean spacing in the bulk: above it the resolvent sees the law, below it the sample.

3.13 B With $z = \lambda + i\varepsilon$,

$$\frac{1}{\lambda_i - z} = \frac{1}{(\lambda_i - \lambda) - i\varepsilon} = \frac{(\lambda_i - \lambda) + i\varepsilon}{(\lambda_i - \lambda)^2 + \varepsilon^2},$$

whose imaginary part is $\frac{\varepsilon}{(\lambda_i - \lambda)^2 + \varepsilon^2} > 0$. Averaging over i gives the display, and positivity of every term gives $\Im m_N > 0$ on the upper half plane. Multiplying by ε ,

$$\varepsilon \Im m_N(\lambda + i\varepsilon) = \frac{1}{N} \sum_{i=1}^N \frac{\varepsilon^2}{(\lambda_i - \lambda)^2 + \varepsilon^2},$$

a finite sum in which each term equals 1 when $\lambda_i = \lambda$ and tends to 0 otherwise. So the limit is $\frac{1}{N}$ times the number of i with $\lambda_i = \lambda$; no convergence theorem is needed, the sum has N terms.

For the picture: the i th term of $\Im m_N$ is a bump centered at λ_i of height $\frac{1}{N\varepsilon}$ and width ε , so $\frac{1}{\pi} \Im m_N(\lambda + i\varepsilon)$ is roughly $\frac{1}{N\varepsilon}$ times the number of eigenvalues within ε of λ . In the bulk that number is about $N\rho_{sc}(\lambda)\varepsilon$, which is a stable average only when $N\varepsilon \gg 1$. For $\varepsilon \ll 1/N$ the nearest eigenvalue dominates and the graph is a comb of spikes of height $\frac{1}{\pi N\varepsilon}$. That is the crossover of Exercise 3.12.

§4 The semicircle law by the Stieltjes transform

4.22 A $A^{-1} = \frac{1}{\alpha d - b^2} \begin{pmatrix} d & -b \\ -b & \alpha \end{pmatrix}$. The corner is $\frac{d}{\alpha d - b^2} = \frac{1}{\alpha - b^2/d}$. The trace is $\frac{\alpha + d}{\alpha d - b^2}$, and the right side of the trace identity is

$$\frac{1}{d} + \frac{d}{\alpha d - b^2} \left(1 + \frac{b^2}{d^2}\right) = \frac{(\alpha d - b^2) + (d^2 + b^2)}{d(\alpha d - b^2)} = \frac{\alpha + d}{\alpha d - b^2}.$$

For $A = M - zI$ with M symmetric 2×2 , $g = (A^{-1})_{11} = (M_{11} - z - \frac{M_{12}^2}{M_{22} - z})^{-1}$ and $\text{tr } A^{-1} - \frac{1}{M_{22} - z} = g(1 + \frac{M_{12}^2}{(M_{22} - z)^2})$. With $t = \Im \frac{M_{12}^2}{M_{22} - z} = \frac{M_{12}^2 \Im z}{|M_{22} - z|^2} \geq 0$, the denominator of g has imaginary part $-\Im z - t$, so $|g| \leq \frac{1}{\Im z + t}$, while $|1 + \frac{M_{12}^2}{(M_{22} - z)^2}| \leq 1 + \frac{M_{12}^2}{|M_{22} - z|^2} = 1 + \frac{t}{\Im z}$. The product is at most $\frac{1+t/\Im z}{\Im z + t} = \frac{1}{\Im z}$, which is (14) for $N = 2$.

4.23 A If λ_i are the eigenvalues of M then $\sigma \lambda_i$ are those of σM , so the transform of σM is $\frac{1}{N} \sum_i \frac{1}{\sigma \lambda_i - z} = \frac{1}{\sigma} m_N(z/\sigma)$. Since $\Im(z/\sigma) = \Im z/\sigma > 0$, Theorem 4.1(b) applies at the point z/σ and gives $\mathbb{E}[\frac{1}{\sigma} m_N(z/\sigma)] \rightarrow \frac{1}{\sigma} m(z/\sigma) =: m_\sigma(z)$, uniformly on bounded sets of $\Re z$. Put $w = z/\sigma$ and $m(w) = \sigma m_\sigma(z)$ into $m(w)^2 + wm(w) + 1 = 0$: $\sigma^2 m_\sigma^2 + z m_\sigma + 1 = 0$. And $|m_\sigma| = |m|/\sigma < 1/\sigma$. Solving, $m_\sigma(z) = \frac{-z + \sqrt{z^2 - 4\sigma^2}}{2\sigma^2}$ with the root of positive imaginary part.

For the density, on the real axis the coefficients are real, so if $\Im m_\sigma \neq 0$ the two roots are conjugate and their product $1/\sigma^2$ gives $|\sigma m_\sigma| = 1$. Write $\sigma m_\sigma = e^{i\theta}$; dividing $\sigma^2 m_\sigma^2 + z m_\sigma + 1 = 0$ by $\sigma e^{i\theta}$ gives $\sigma e^{i\theta} + x + \sigma e^{-i\theta} = 0$, so $x = -2\sigma \cos \theta$ and $\Im m_\sigma = \frac{\sin \theta}{\sigma} = \frac{1}{\sigma} \sqrt{1 - \frac{x^2}{4\sigma^2}} = \frac{1}{2\sigma^2} \sqrt{4\sigma^2 - x^2}$. Dividing by π gives the density, supported on $[-2\sigma, 2\sigma]$. Proposition 3.7 then turns this into the proportion of eigenvalues of σM in every interval, exactly as in Theorem 4.19.

4.24 B The independent variables are M_{ij} , $i \leq j$. Since $f = \sum_{i,j} M_{ij}^2 = \sum_i M_{ii}^2 + 2 \sum_{i < j} M_{ij}^2$, we get $\partial f / \partial M_{ij} = 4M_{ij}$ for $i < j$ and $\partial f / \partial M_{ii} = 2M_{ii}$. The Poincaré sum is

$$\sum_{i < j} \frac{1}{N} \mathbb{E}(4M_{ij})^2 + \sum_i \frac{2}{N} \mathbb{E}(2M_{ii})^2 = \frac{16}{N} \binom{N}{2} \frac{1}{N} + \frac{8}{N} \cdot N \cdot \frac{2}{N} = 8 \left(1 - \frac{1}{N}\right) + \frac{16}{N} = 8 + \frac{8}{N},$$

and Lemma 4.2 multiplies it by $\frac{\pi^2}{8}$. The bound is $O(1)$ although $\text{tr } M^2$ is a sum of N^2 terms; the exact variance $4 + \frac{4}{N}$ is smaller by the factor $\frac{\pi^2}{2}$ and the crude constant. What Poincaré sees is that each of the N^2 partial derivatives is of size $N^{-1/2}$ and their squares sum to $O(1)$ — the same mechanism as in Proposition 4.3, with M in place of R^2 .